

Assignment 7

Assignment Title

Design and Develop a Full Stack Web Application using React.js, Node.js (Express), and MongoDB

(Student Feedback Review System)

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Objectives

- To design a full-stack web application using React.js for frontend
 - To develop RESTful APIs using Node.js and Express
 - To integrate MongoDB for storing feedback data
 - To implement CRUD operations for student feedback
 - To understand client-server architecture
 - To build a dynamic and responsive user interface
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Outcomes

- Learned full-stack development using React, Node.js, and MongoDB
 - Understood API integration between frontend and backend
 - Implemented dynamic UI using React components
 - Performed CRUD operations using MongoDB
 - Gained experience in state management and API calls
 - Improved debugging and real-world development skills
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Theory Concept in Brief

1. Frontend (React.js)

The frontend is developed using React.js, which provides a component-based architecture.

Features:

- Dynamic UI rendering
- Form handling for feedback submission
- State management using hooks
- API integration using fetch/axios

Users can:

- Submit feedback
- Filter feedback by course
- View recent feedback
- Delete feedback

2. Backend (Node.js + Express.js)

The backend handles API requests and business logic.

Features:

- REST API creation
- Routing for feedback operations
- Middleware for JSON parsing
- Communication with MongoDB

Example Routes:

- /api/feedback → Add feedback
- /api/feedback → Get all feedback
- /api/feedback/:id → Delete feedback

3. Database (MongoDB)

MongoDB is used as a NoSQL database.

Stored Data:

- Student Name
- Course
- Rating
- Comments
- Timestamp

Mongoose is used to define schemas and interact with MongoDB.

Technologies Used

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB
 - Mongoose
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Core Functionalities Implemented

Student Feedback System

- Add Feedback (Name, Course, Rating, Comments)
- View Feedback List
- Delete Feedback
- Filter Feedback by Course
- Display Date and Time
- Responsive UI

Features from Your Project UI

(From your screenshots)

- Dashboard titled: "**Student Feedback Review**"
- Welcome message with user name
- Filter option for course (e.g., Computer Engineering)
- Feedback submission form with:
 - Student Name
 - Course
 - Rating (Dropdown)
 - Comments
- Feedback display section showing:
 - Name and Course
 - Rating (e.g., 5/5)
 - Comments
 - Timestamp
 - Delete button

Methodology / Steps Followed

1. Designed frontend UI using React components
2. Created Express server using Node.js
3. Connected MongoDB using Mongoose
4. Defined Feedback schema
5. Implemented REST APIs
6. Integrated frontend with backend using API calls
7. Implemented CRUD operations
8. Added filtering functionality
9. Tested application

10. Debugged errors and improved UI

Program Code (Overview)

Backend Example

```
app.use('/api/feedback', feedbackRoutes);
```

Feedback Schema Example

```
const feedbackSchema = new mongoose.Schema({  
  name: String,  
  course: String,  
  rating: Number,  
  comment: String,  
  createdAt: {  
    type: Date,  
    default: Date.now  
  }  
});
```

Real-Life Applications

- College feedback systems
 - Course evaluation platforms
 - Employee feedback systems
 - Product review systems
 - Survey applications
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Output Screenshots

Include:

- Homepage (Feedback Dashboard)
- Feedback Form
- Feedback List
- MongoDB Data (Compass Screenshot)

From your UI:

- Clean dashboard layout
- Form-based input system
- Feedback cards with delete option
- Filter functionality

Student Feedback Review

Collect student course feedback, ratings, and comments.

Welcome, YASH GAJANAN MAHAJAN. You can now add or remove feedback.
 Sign out

Filter by course

Clear

Submit Feedback

Student Name

Course

Rating

Comments

Save Feedback

Recent Feedback

yash — comp
 Delete

Rating: 5 / 5

noce

14/4/2026, 7:09:53 pm

MongoDB Compass - TY_FSDL/student-feedback/feedbacks

Connections Edit View Collection Help

Compass

My Queries
 Data Modeling

CONNECTIONS (3)

TY_FSDL
 ASS6_2
 ASS6
 ASS7
 students feedback
 Poocoe
 admin
 college_website
 config
 eventDB
 local
 loginDB
 marketplace
 poocoe
 practice
 student-feedback
 feedbacks
 users
 taact

students feedback
 feedbacks

TY_FSDL > student-feedback > feedbacks

Documents 2
 Aggregations
 Schema
 Indexes 1
 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain
 Reset
 Find
 Options

ADD DATA
 UPDATE
 DELETE
 EXPORT DATA
 EXPORT CODE

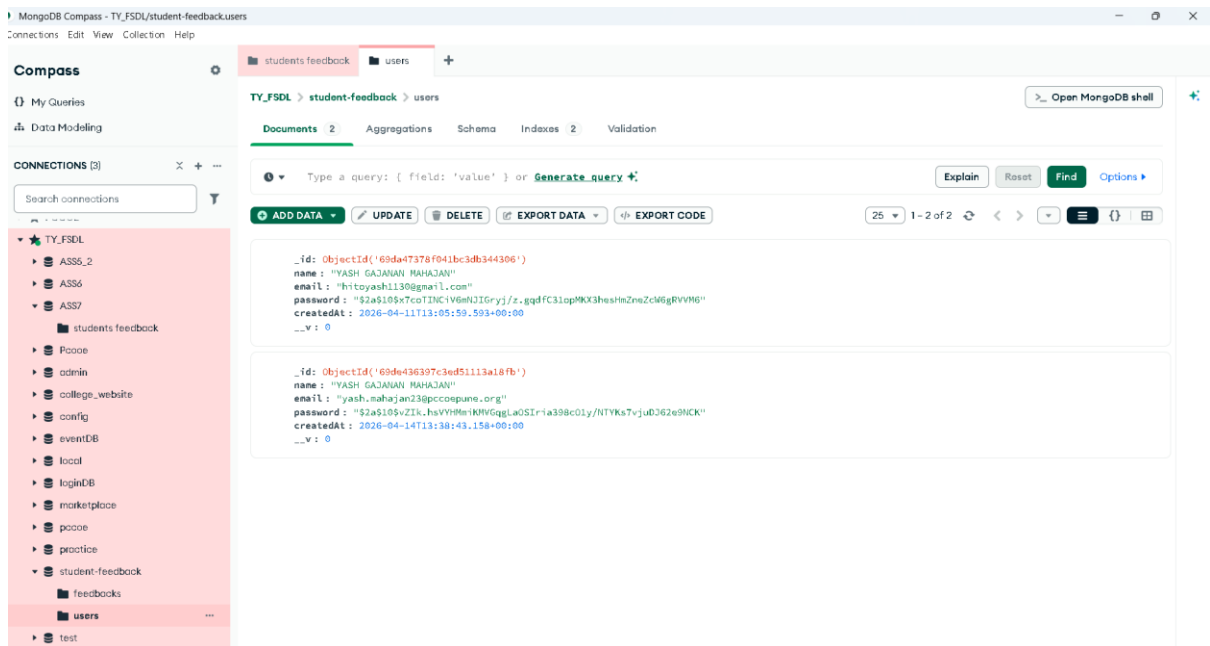
25
 1 - 2 of 2
 < >
 []
 []
 []

```

_id: ObjectId('60da7538f041bc3db34431c')
studentName: 'yash'
course: 'comp'
rating: 3
comments: 'new syllabus required'
createdAt: 2026-04-11T13:06:27.299+00:00
__v: 0

_id: ObjectId('60da436d97c3ed5113a18fd')
studentName: 'yash'
course: 'comp'
rating: 5
comments: 'noce'
createdAt: 2026-04-14T13:38:53.918+00:00
__v: 0

```



Conclusion

The assignment successfully demonstrates the development of a full-stack web application using React.js, Node.js, Express, and MongoDB.

The Student Feedback Review System allows users to submit, view, filter, and delete feedback efficiently.

This project provides hands-on experience in frontend-backend integration, database handling, and API development, helping in building scalable real-world applications.